CH 12 S2

11. **Professor Asimov** Professor Isaac Asimov wrote nearly 500 books during a 40-year career. In fact, as his career progressed, he became even more productive in terms of the number of books written within a given period of time.[3] The data give the time in months required to write his books in increments of 100:

Number of Books, x	100	200	300	400	490
Time in Months, y	237	350	419	465	507

a. Assume that the number of books x and the time in months y are linearly related. Find the least-squares line relating y to x.

b. Plot the time as a function of the number of books written using a scatterplot, and graph the least-squares line on the same paper. Does it seem to provide a good fit to the data points?

c. Construct the ANOVA table for the linear regression.

a. $\hat{y} = a + bx$

$b = \frac{S_{xy}}{S_{xx}} = \frac{64386}{96080} = 0.670 \quad a = \bar{y} - b\bar{x} = 195.94$

$S_{xy} = \sum xy - n\bar{x}\bar{y} = 653830 - 5 \times 298 \times 395.6 = 64386$

$S_{xx} = \sum x^2 - n\bar{x}^2 = 540100 - 5 \times 88804 = 96080$

$\hat{y} = 195.94 + 0.67x$

c.

$SS = S_{yy} = \sum y^2 - n\bar{y}^2 = 45007.2$

$SSR = \frac{S_{xy}^2}{S_{xx}} = \frac{(\sum xy - n\bar{x}\bar{y})^2}{\sum x^2 - n\bar{x}^2} = 43146.929$

$SSE = SS - SSR = 1860.27$

ANOVA	SS	df	MS	=
SSR	43146.929	1	43146.929	69.5817
SSE	1860.27	3	620.09	
SS	45007.2	4		